Duty at home

DM10 · Points of a tetrahedron · 50 min.

A tetrahedron is considered $ABCD$ not flattened. Points I, J and K are the respective circles of $[AB]$, $[AC]$ and $[AD]$.

1. Express $\overline{IJ}$ according to the $\overrightarrow{BC}$ and $\overrightarrow{IK}$ according to the $\overrightarrow{BD}$.
2. Deducing that plans (IJK) and (BCD) are parallel.
3. Define point G by $\overrightarrow{AG} = \frac{1}{4}(\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD})$. The point G Does it belong to the plan? (IJK) ? Justify.
4. For this question alone, it is assumed that, in an orthonormal coordinate system, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$ and $D(0;0;4)$. Calculate lengths IJ, IK and JK.